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Ambient Systems: Company Background

- Incorporated in 2021 in Singapore (2024 in Canada)

  -off from National University of Singapore (NUS)
- Received investment of $100k from NUS deep tech investment arm
- Received awards and recognition from Google Cloud and Panasonic for its product and innovation
- Seeking to develop MVP for commercialization

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Project Objective

- Develop MVP for four industries
- Create a demo for four industries
- Demonstrate real-time data collection and visualization capability using off-the-shelf sensor and cloud IoT service

  current MVP, cloud-based software, machine learning ardware, and IP

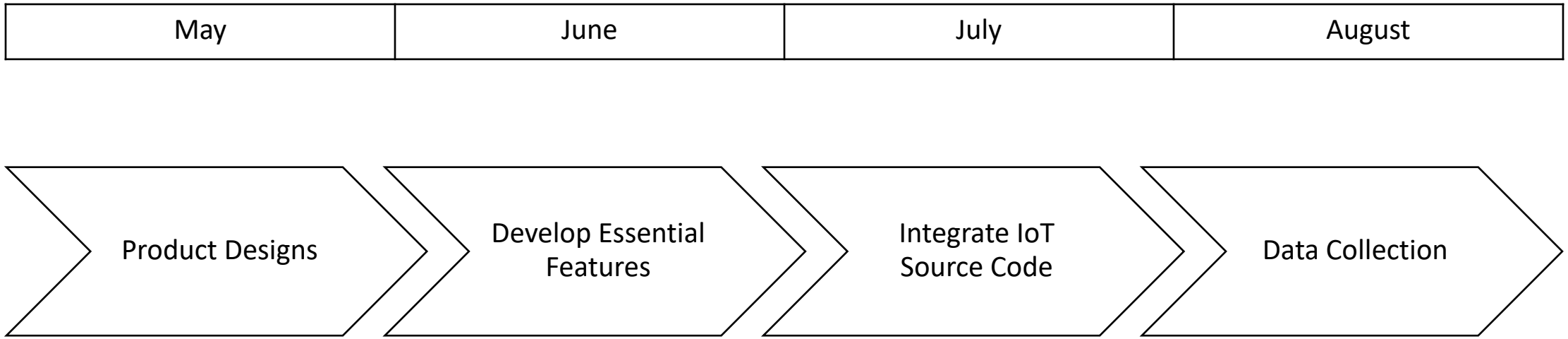
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Current Status

	Use Case #1: Vertical Farming	Use Case #2: Decarbonization Platform for Commercial Real Estate	Use Case #3: Transportation
Status	Successfully developed a proof-of-concept for vertical farming in Singapore, incorporating multi-modal sensors and machine learning and secured an exclusive patent license for the invention.	Won first runner-up at a Google Cloud hackathon in Singapore for our innovative cloud-based AI solution for decarbonization in the commercial real estate sector.	Emerged as the winner of 2023 Panasonic Deep Tech Innovation Challenge, paving the way for a collaboration to leverage Panasonic's fast data transfer SW and 60 GHz V2X expertise in the transportation industry.
Components	• Hardware Sensors and Network • ML model and patent • Proprietary data	• Cloud-based platform • ML model • Open Source Academic Data	• System diagram

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Project Charter: MVP Timeline



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User Personas

			ona
Role	Usuall lanning and forecasting.	Usual for costs and compliance.	Overs ng fleet mana gement, routing, and logistics.
Responsibilities	Focuses on financial planning and analysis, including forecasting revenues and expenses, budgeting, and providing insights to support strategic decision-making.	Primarily concerned with managing costs, ensuring compliance with regulatory requirements and internal policies, and maintaining accurate financial records.	Ensures efficient vehicle and driver utilization while managing routing and scheduling to minimize transit times and costs, and coordinates with suppliers to optimize freight movement.
Needs	Requires tools and features that enable accurate forecasting, scenario planning, budget management, and the ability to analyze financial data to support business objectives.	Requires tools and features that streamline cost management processes, facilitate compliance with regulatory standards, automate financial reporting tasks, and provide visibility into financial data for auditing and reporting purposes.	Requires real-time tracking and monitoring tools for fleet management, alongside insights for route optimization, fuel savings, and operational efficiency.

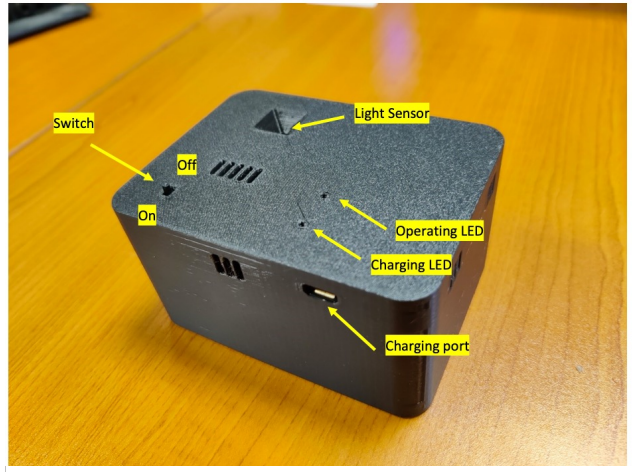
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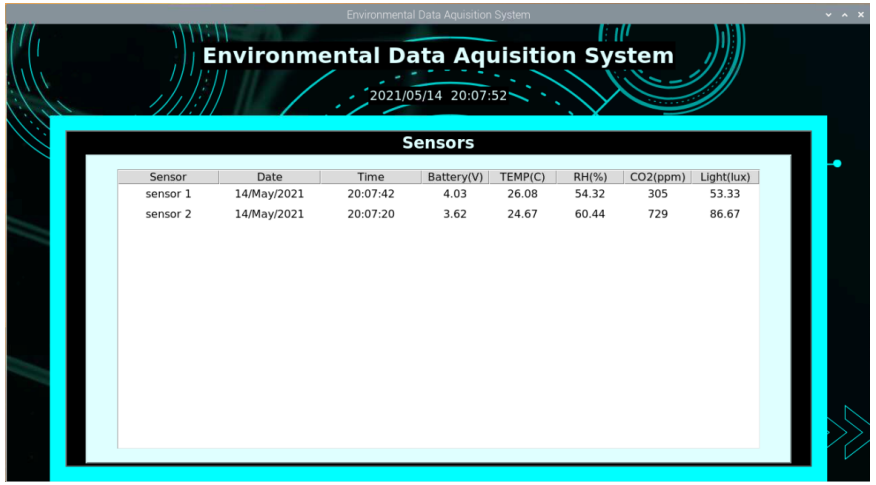
Project Charter: User Stories

Finance Teams	Ops Teams	Ops and Finance	Data
• Finance teams can forecast revenues and expenses. • Finance teams can prepare reports for securing long-term funding and investments. • Finance teams can calculate and minimize tax. • Finance teams can reduce costs. • Finance teams can manage cash flows and revenues transparently • Finance teams can identify and manage risks. • Finance teams can ensure compliance with regulations. • Finance teams can analyze financial data to for decision-making.	• Operations teams can translate metrics/goals/KPIs into operational tasks. • Operations teams can map operational tasks into financial goals. • Operations teams can identify operational drivers. • Operations teams can prioritize and sequence key value-added tasks. • Operations teams can define task objectives and success criteria. • Operations teams can use data to communicate and align.	• Ops and finance teams can align goals and objectives between finance and operations. • Ops and finance teams can see the relevant data for each of our work tasks. • Ops and finance teams can collaborate on analysis and decision making. • Ops and finance teams can do a continuous improvement of the process. • Ops and finance teams can partake in shared accountability and performance management. • Ops and finance teams can utilize communication tools for seamless collaboration.	• Finance teams can access data from different systems. • Finance teams can get data from operations using IoT.

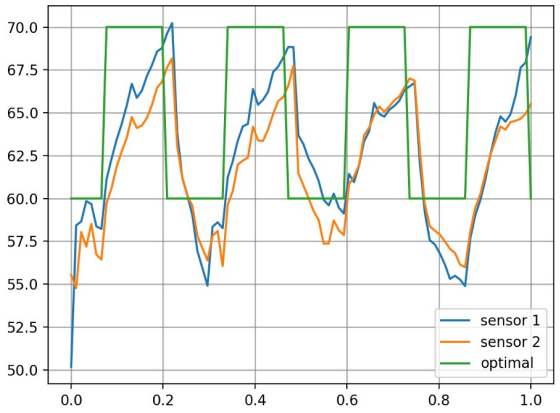
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MVP #1: Vertical Farming





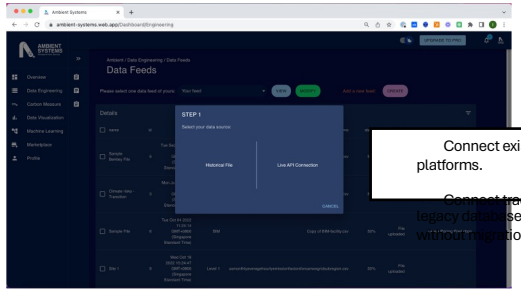






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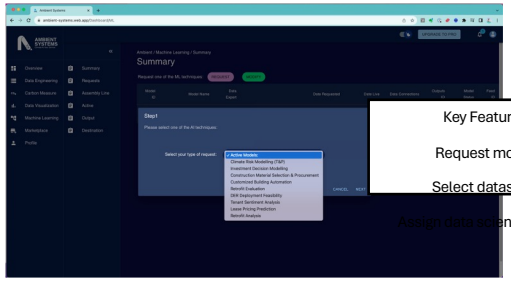
User Journey: Feature 1 – Combine Datasets



Connect existing platforms.

Connect traditional legacy databases without migration.

User Journey: Feature 4 – Leverage ML and AI



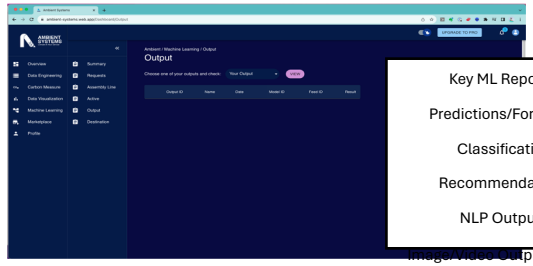
Key Features

Request model

Select datasets.

Assign data science expert

User Journey: Step 5 – Output



Key ML Reports:

Predictions/Forecasts

Classification

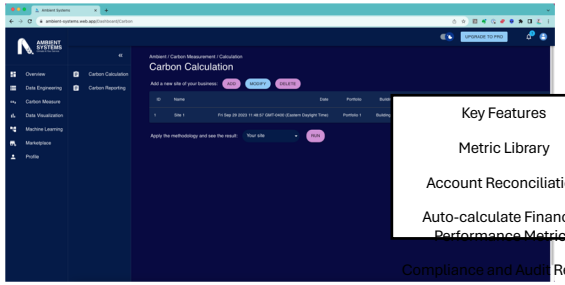
Recommendations

NLP Outputs

Image Class Output: Object detection, facial recognition, image captioning

Explanations/Interpretations

User Journey: Feature 2 – Automate Workflow



Key Features

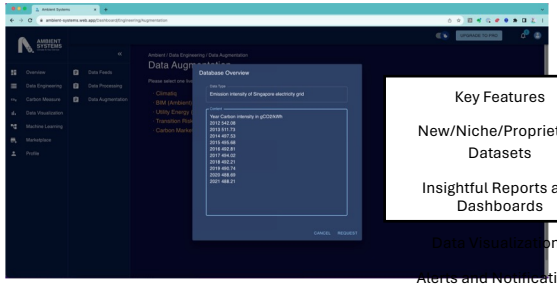
Metric Library

Account Reconciliation

Auto-calculate Financial Performance Metrics

Compliance and Audit Reports

User Journey: Feature 3 – Add New Data



Key Features

New/Niche/Proprietary Datasets

Insightful Reports and Dashboards

Data Visualizations

Alerts and Notifications

3/22/24

MVP #2: Commercial Real Estate

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MVP #3: Transportation

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MVP #4: Construction

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Develop Essential Features

- Migration of for the application: GCP to OVH

  ML Pipeline: GCP (vertex)
- Demonstrate plug-and-play data sensor to server (AWS/GCP) in real-time
- Data Governance Feature (managing data access and model requests)

  ETL
- Import metrics library for three industries
- Import data for four industries demos (vertical farms, CRE, construction, transportation bus fleets

  continuous delivery pipeline and documentation (i.e., JIRA, GitHub, , user management, quality testing)

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